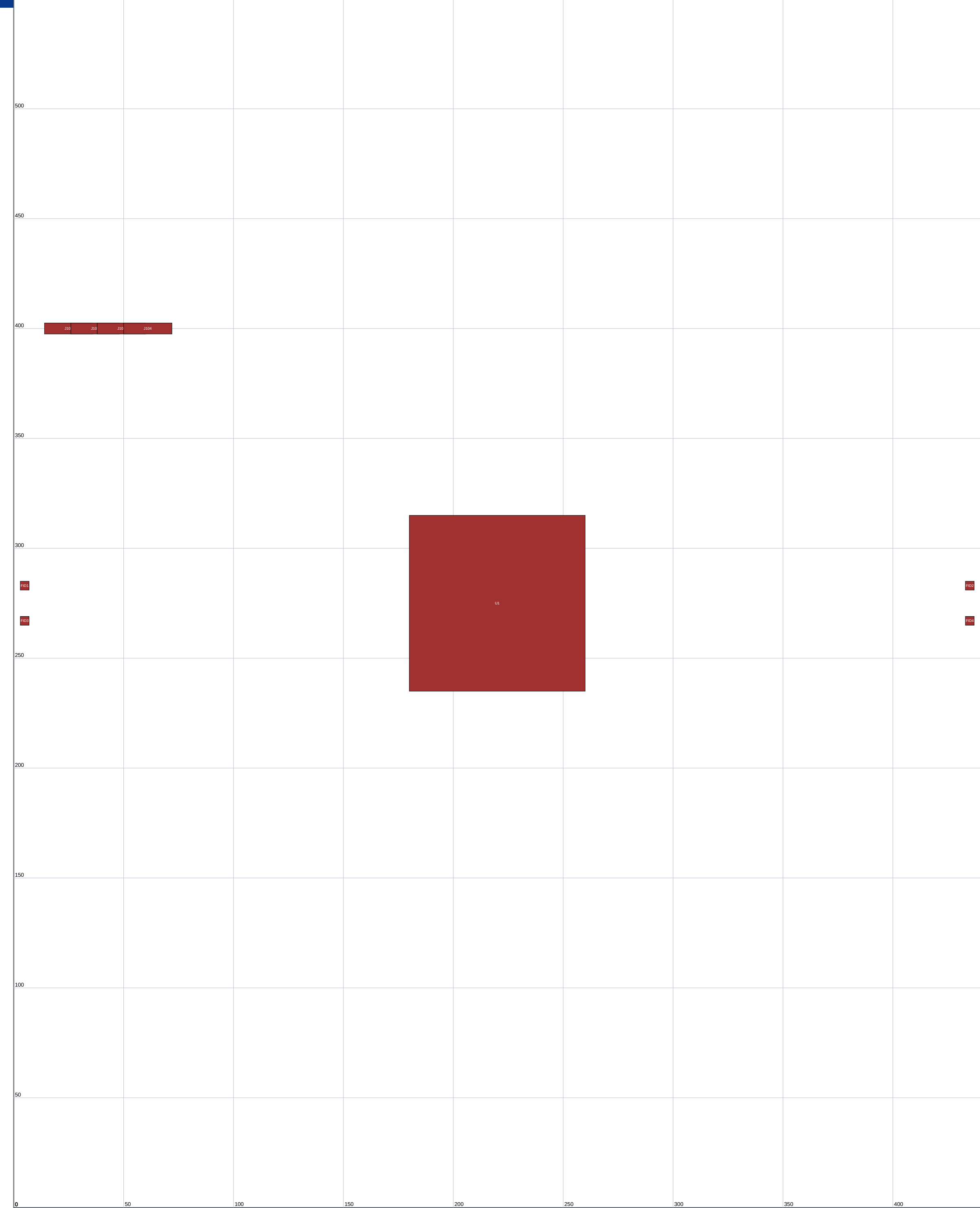


Layer characteristics

Layer name	In13.Cu
Stack-up index	14 of 27
Role	Signal
Net / signal	Mixed
Copper weight	2 oz
Dielectric above	Megtron-7 · Dk 3.40 · Df 0.002 @ 10 GHz
Reference layer	GND
Impedance target	100 Ω diff · 50 Ω SE ±5 %
Antipad rule	0.20 mm clearance · BGA escape Type-A
Microvia type	IPC-4761 Type VII · filled-and-capped
Back-drill	≤ 6 mil stub · PCIe Gen 6 / HBM4
Trace width	4.0 mil / 4.0 mil clearance
Test coupon	IPC-2141 Appendix B
AOI inspection	100 % bare-board ET + AOI
Fab note ref	See Fab_Drawing notes 1–18

Compliance summary

- IPC-6012 Class 3/A acceptance, IPC-A-600 Class 3 visual
- IPC-2581 Rev C unified data package included
- IPC-7351B 2D-data-matrix date code on B.Silk
- IPC/JEDEC J-STD-033 MSL-3 packaging
- RoHS / REACH compliant · Pb-free SAC305 reflow
- Conformal coating: none (sealed liquid-cooled chassis)

LightRail AI — Confidential / Investor data-room	
Programme	LR-S1A 1U CPO Switch Motherboard
Drawing No.	LR-S1A-LAY-014
Layer	In13.Cu
Role	Signal
Designer	LightRail AI — Design Automation
Date / Rev	2026-04-19 · Rev 1.0